

# JS Operators

## 1. What is Operator and Operand?

- a. **Operator** → An operator is a symbol or function that performs a specific operation on one or more operands.
- b. **Operand** → An operand is a value or variable on which the operator performs operations.
- c. **For Example:**  
let a = 5 + 3;  
**Operator:** + (addition)  
**Operands:** 5 and 3

## 2. Types of Operators:

- a. **Operators are categorized into the following types:**
  - i. Unary Operator
  - ii. Arithmetic Operator
  - iii. Assignment Operator
  - iv. Comparison Operator
  - v. Logical Operator
  - vi. Ternary Operator
  - vii. Comma Operator
  - viii. String Operator
  - ix. Nullish Coalescing Operator (??)

## 3. Unary Operator:

- a. These operators work on a single operand.
- b. **Types:**

- i. **Unary Plus (+):**
  - 1. If the value is already a number → nothing changes.
  - 2. If it's a string containing a number → it gets converted to a number.
- ii. **Unary Minus (-):**
  - 1. Negates the value of an operand.
- iii. **Increment (++):**
  - 1. Increases the value by 1.
  - 2. Pre-Increment (++value)
  - 3. Post-Increment (value++)
- iv. **Decrement (--):**
  - 1. Decreases the value by 1.
  - 2. Pre-Decrement (--value)
  - 3. Post-Decrement (value--)
- v. **Logical Not (!):** Inverts a boolean value.
- vi. **For Example:**

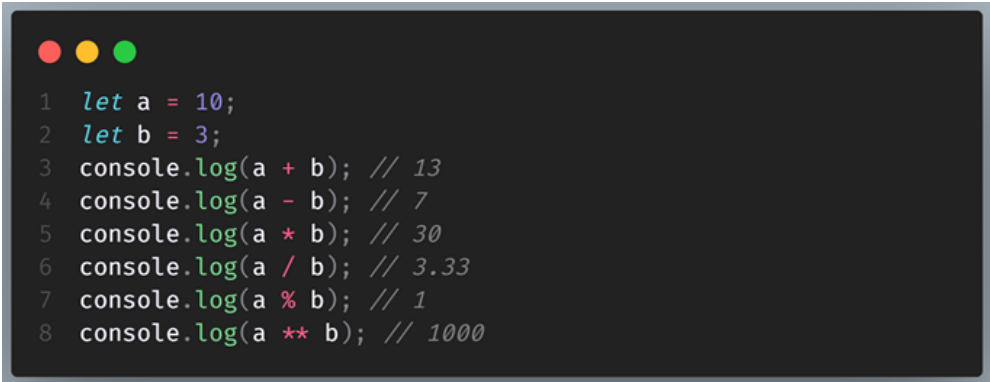
```
1 let a = 5;
2 console.log(+a); // 5
3 console.log(-a); // -5
4 console.log(++a); // 6 (Pre-Increment Operator)
5 console.log(--a); // 5 (Pre-Decrement Operator)
6 console.log(a++); // 5 (Post-Increment Operator)
7 console.log(a--); // 6 (Post-Decrement Operator)
8 console.log(!true); // false
```

#### 4. Arithmetic Operator:

a. These operators perform basic mathematical operations.

b. **Types:**

- i. *Addition (+)*
- ii. *Subtraction (-)*
- iii. *Multiplication (\*)*
- iv. *Division (/)*
- v. *Modulus (%)*: Remainder of division.
- vi. *Exponentiation (\*\*)*: Raises one number to the power of another.
- vii. **For Example:**



```
1 let a = 10;
2 let b = 3;
3 console.log(a + b); // 13
4 console.log(a - b); // 7
5 console.log(a * b); // 30
6 console.log(a / b); // 3.33
7 console.log(a % b); // 1
8 console.log(a ** b); // 1000
```

#### 5. Assignment Operator:

a. These operators assign values to variables.

b. **Types:**

- i. *Simple Assignment (=)*: Assigns the value.
- ii. *Addition Assignment (+=)*: Adds and assigns.
- iii. *Subtraction Assignment (-=)*: Subtracts and assigns.
- iv. *Multiplication Assignment (\*=)*: Multiplies and assigns.
- v. *Division Assignment (/=)*: Divides and assigns.
- vi. *Modulus Assignment (%=)*: Modulus and assigns.

vii. **For Example:**

```
1 let a = 5;
2 a += 3; // a = a + 3
3 console.log(a); // 8
4
5 a *= 2; // a = a * 2
6 console.log(a); // 16
```

## 6. Comparison Operator:

a. These operators compare two values and return a boolean result.

b. Types:

- i. *Equal to ( == )*: It will only check the value not datatype.
- ii. *Strict equal to ( === )*: It will check the value as well as datatype. (**Important**)
- iii. *Not equal to ( != )*:
- iv. *Strict not equal to ( !== )*:
- v. *Greater than ( > )*:
- vi. *Less than ( < )*:
- vii. *Greater than or equal to ( >= )*:
- viii. *Less than or equal to ( <= )*:
- ix. **For Example:**

```
1 console.log(5 == "5"); // true
2 console.log(5 === "5"); // false
3 console.log(10 > 5); // true
4 console.log(10 <= 5); // false
```

## 7. Logical Operator:

a. These operators work with boolean values.

b. **Types:**

- i. **AND (&&):** Returns **true** if **both** operands are **true**.
- ii. **OR (||):** Returns **true** if **at least one** operand is **true**.
- iii. **NOT (!):** Inverts the boolean value.

If the value is truthy → returns **false**.

If the value is falsy → returns **true**.

iv. **For Example:**

```
1 console.log(true && false); // false
2 console.log(true && true); // true
3 console.log(true || false); // true
4 console.log(!true); // false
```

## 8. Ternary Operator / Conditional Operator:

a. It is a shorthand for an **if-else** statement.

b. Syntax: **condition ? value\_if\_true : value\_if\_false**

c. **For Example:**

```
1 let age = 18;
2 let canVote = age ≥ 18 ? "Yes" : "No";
3 console.log(canVote); // Yes
```

## 9. Comma Operator:(,)

- a. Used to evaluate multiple expressions and return the value of the last expression.
- b. For Example:

```
1 let a = (1, 2, 3); // 3
2 console.log(a);
```

## 10. String Concatenation Operator:(+)

- a. Used to concatenate strings.
  - i. *Concatenation (+)*: Combines two strings.
  - ii. *Concatenation Assignment (+=)*: Appends a string.
- b. For Example:

```
1 let firstName = "John";
2 let lastName = "Doe";
3 console.log(firstName + " " + lastName); // John Doe
4
5 let greeting = "Hello";
6 greeting += ", World!";
7 console.log(greeting); // Hello, World!
```

## 11. Nullish Coalescing Operator (??):

- a. The *Nullish Coalescing Operator* is represented by two question marks (??).
- b. It takes two operands and returns the first operand if it is not null or undefined. Otherwise, it returns the second operand.
- c. It is a logical operator introduced in *ES2020*.

d. **Purpose:** This operator is useful when dealing with variables that might hold *null* or *undefined* values and allows you to provide default values to avoid unexpected behavior or errors.

e. **Syntax:**



```
let result = operand1 ?? operand2;
```

f. **operand1:** The value to check if it is not *null* or *undefined*.

g. **operand2:** The fallback or default value to return if *operand1* is *null* or *undefined*.

h. **For Example:** Basic Example

```
1 let name = null;
2 let defaultName = "Guest";
3
4 let result = name ?? defaultName; // Returns "Guest"
5 console.log(result);
```

i. **For Example:** Using with Functions

```
1 function greet(userName) {
2   return userName ?? "Anonymous";
3 }
4
5 console.log(greet("Alice")); // Returns "Alice"
6 console.log(greet(null)); // Returns "Anonymous"
7 console.log(greet(undefined)); // Returns "Anonymous"
```